

Structural Analysis and Optimization of an RC Car Shock Tower

Static Analysis – Modal Analysis – Fatigue Life Assessment – Topology Optimization

Prepared & done by: **ENG-Majdy Al Sous**
Mechanical Design Engineer
Majdy32047@gmail.com
[LinkedIn](#)

Software used : *SOLIDWORKS-CAD & simulation*

1. Objective

The objective of this project is to evaluate the structural performance of an RC vehicle shock tower using finite element analysis techniques.

The study includes:

- Static structural analysis
- Modal (frequency) analysis
- Fatigue life evaluation
- Topology optimization

The goal is to improve the structural performance while reducing unnecessary material.

1. Reference Vehicle

The shock tower analyzed in this project is based on the **LiteHawk-NOMAND RC vehicle**.

The component geometry was reconstructed using reference images and publicly available specifications from the manufacturer. `



Figure 1. shock tower vehicle used as the reference for the analysis.



Figure 2. LiteHawk NOMAND RC Original shock tower mounted on the vehicle suspension assembly.

2. Geometry Reconstruction

The shock tower geometry was recreated in SolidWorks using reverse engineering methods based on visual references and dimensional approximations.

This allowed the creation of a CAD model suitable for structural simulation and design evaluation.

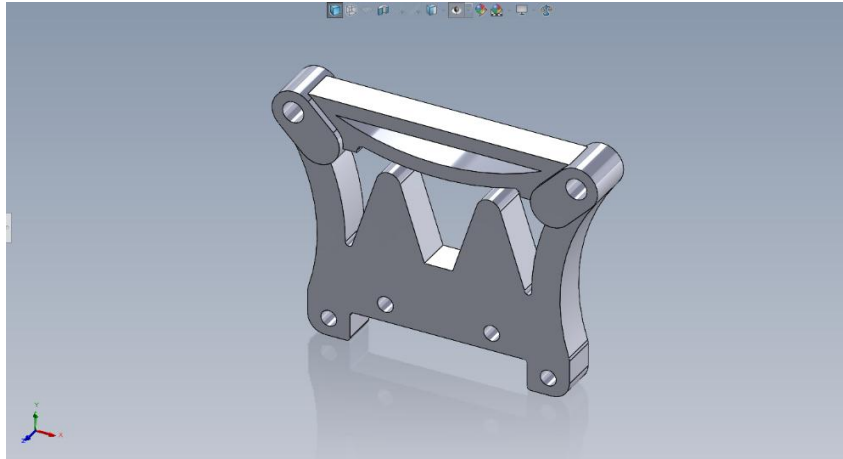


Figure 3. Reconstructed CAD model of the shock tower created in SolidWorks.

3. Material Selection and Assumptions

Initially, the shock tower was intended to be manufactured from **ABS plastic**, which is commonly used in RC vehicles due to its lightweight and impact resistance.

Static and modal analyses were performed using ABS material properties.

However, fatigue analysis requires **S-N curves based on ASME fatigue data**, and ABS was not available in the SolidWorks fatigue database.

Therefore, the material was changed to **Aluminum 1060 Alloy** for the fatigue and topology optimization studies.

Table 1. material summery used in each study.

Study	Material
Static Analysis	ABS
Modal Analysis	ABS
Fatigue Analysis	Aluminum 1060
Topology Optimization	Aluminum 1060

The first configuration represents the **original solid plate geometry** with no internal cutouts or reinforcements.

Simulation results showed:

- **Maximum Stress:** 2.56 MPa
- **Maximum Displacement:** 0.069 mm

This configuration was used as the **reference design** for comparison with the modified versions.

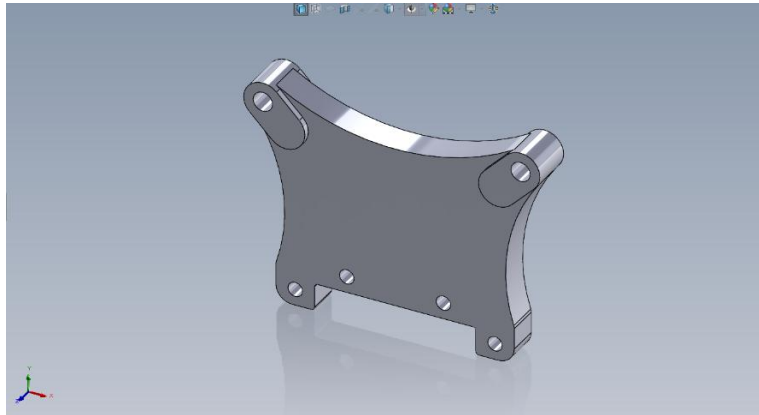


Figure 5. Baseline shock tower configuration used as the reference model.

- ***Configuration 2 – Lightweight Design***

The second configuration introduced **internal material removal** to reduce overall mass.

While this modification successfully reduced weight, it had a noticeable effect on structural stiffness.

Key observations:

- Mass decreased by approximately **22%**
- **Stress increased significantly**
- **Displacement increased considerably**

This result indicates that excessive material removal can weaken load paths and reduce structural stiffness.

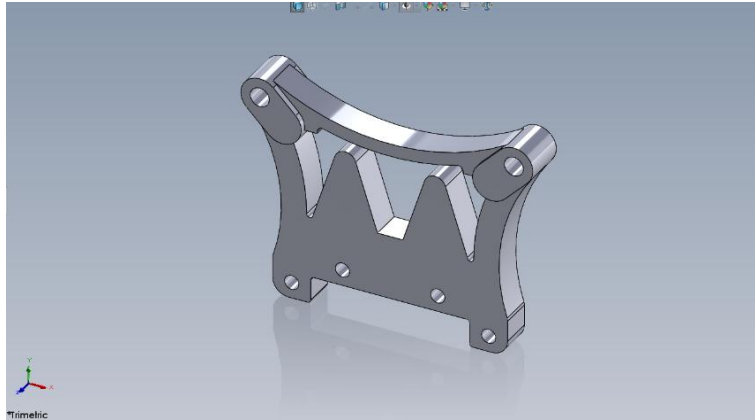


Figure 6. Lightweight shock tower configuration with internal material removal.

- *Configuration 3 – Reinforced Lightweight Design*

The third configuration introduced **internal cutouts combined with a reinforcing bridge between the two pin mounting locations.**

This design aimed to maintain reduced mass while improving stiffness.

Simulation results showed:

- Mass remained **lower than the baseline design**
- **Maximum displacement reduced to 0.055 mm**
- The structure became noticeably **stiffer**

This configuration demonstrated a better balance between **weight reduction and structural rigidity.**

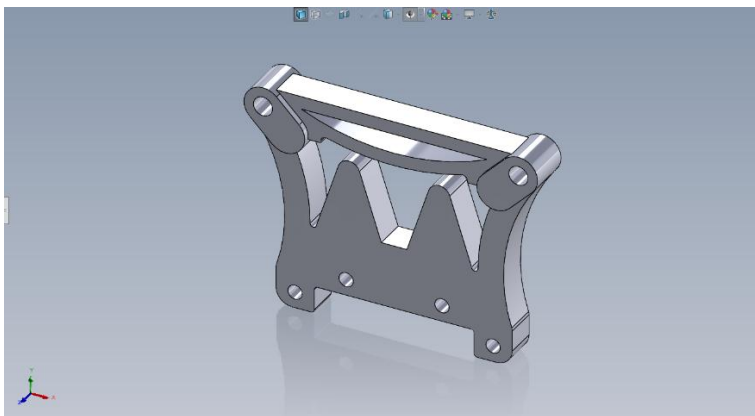


Figure 7. Reinforced lightweight configuration with an internal bridge between the mounting pins.

8. Design Trend Analysis

A **trend tracker study** was conducted to compare the structural performance of the three configurations.

The following parameters were monitored:

- Mass
- Maximum stress
- Maximum displacement (Y-direction)

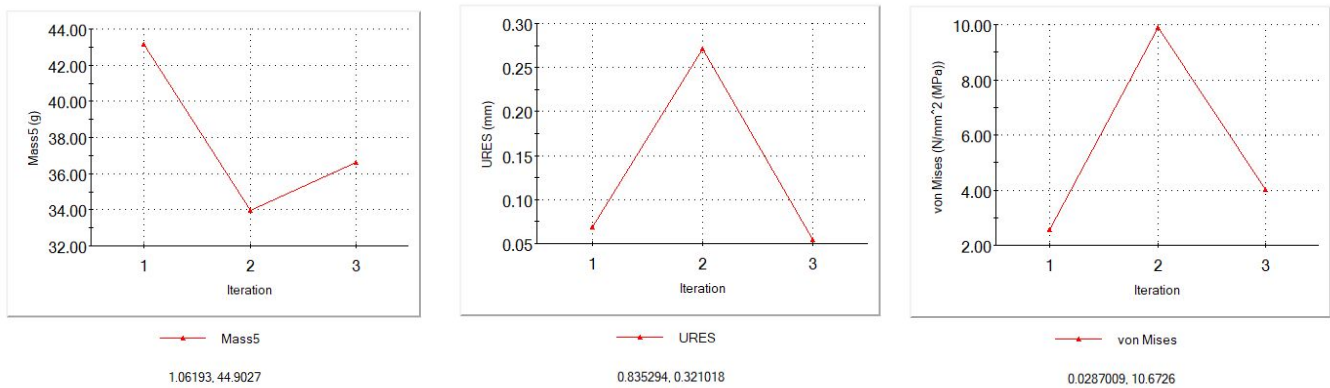


Figure 8. Trend tracker results comparing mass, stress, and displacement across the three design configurations.

9. Structural Behavior Observation

An interesting observation appeared during the analysis of the third configuration.

Although the structure became **stiffer and displacement decreased**, the **maximum stress increased compared to the baseline design**.

At first glance this may seem counterintuitive.

However, from a mechanics perspective this behavior can be explained by **load path redistribution**.

Increasing stiffness alters how loads travel through the structure. While the overall deformation may decrease, stresses can become more concentrated in specific regions where loads are transferred.

This phenomenon is common in structural optimization and highlights the importance of evaluating **both stiffness and stress simultaneously** when modifying geometry.

Table 2. Numerical summary of mass, maximum stress, and displacement for the three trend tracker configurations.

Configuration	Mass (kg)	Max Stress (MPa)	Max Displacement (mm)
Baseline	43.1411	2.5568	0.06887
Lightweight	33.9774	9.9003	0.2713
Reinforced	36.6372	4.0142	0.05496

10. Frequency Analysis

To verify the natural frequencies of the shock tower against excitation frequencies generated by the vehicle's motor and drivetrain, and assess potential resonance risks.

1. Motor Excitation Frequencies

- Motor: 380 brushed motor (RC car LiteHawk IGNITER)
- Estimated RPM under load: 18,000 RPM
- Fundamental motor frequency ($1X$):

$$f = \frac{RPM}{60} = \frac{18000}{60} = 300$$

Harmonic	Frequency (Hz)
1X	300
2X	600
3X	900
4X	1200
5X	1500

2. Shock Tower Natural Frequencies (from the simulation)

Table 2. modes numbers with their respective frequencies

Mode	Natural Frequency
Mode 1	423 Hz
Mode 2	569 Hz
Mode 3	589 Hz
Mode 4	1423 Hz
Mode 5	1597 Hz

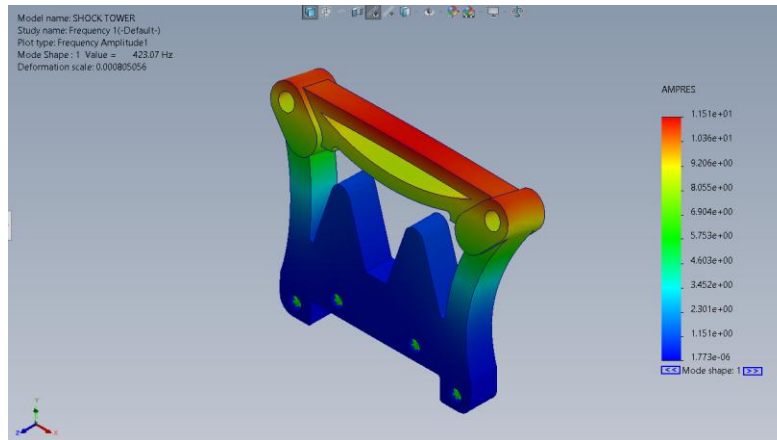


Figure 8. First vibration mode shape of the shock tower.

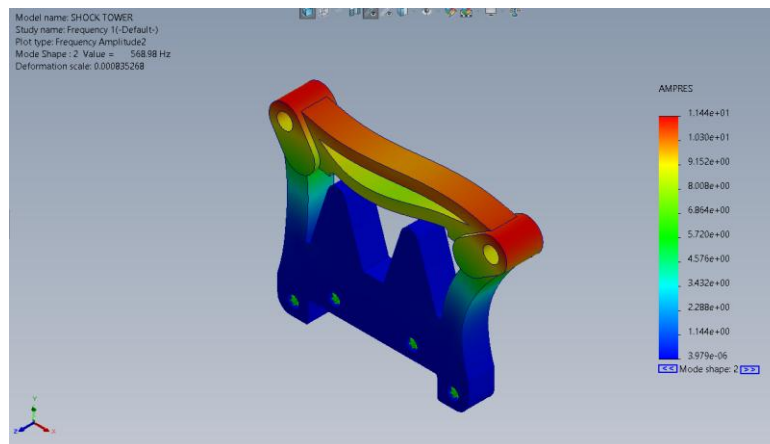


Figure 9. second vibration mode shape of the shock tower.

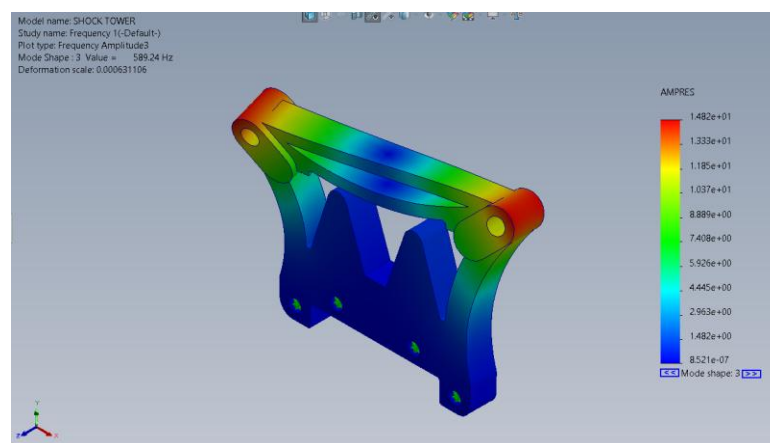


Figure 10. third vibration mode shape of the shock tower.

3. Resonance Assessment

- The first three modes are most critical.
- Mode 2 & 3 (569-589 Hz) are very close to the 2nd motor harmonic (600 Hz), only ~1.8% difference which means **high risk of resonance**.
- Mode 1 (423 Hz) is above fundamental motor frequency (300 Hz) , acceptable.
- Mode 4 & 5 (>1400 Hz) are sufficiently distant from excitation frequencies , considered safe.

11. Fatigue Analysis

A fatigue study was conducted to evaluate the durability of the shock tower under cyclic loading conditions.

Simulation parameters:

- Loading condition: **Fully reversed**
- Total cycles: **500,000 cycles**
- Material: **Aluminum 1060 Alloy**

The fatigue analysis predicts structural life and potential damage accumulation.

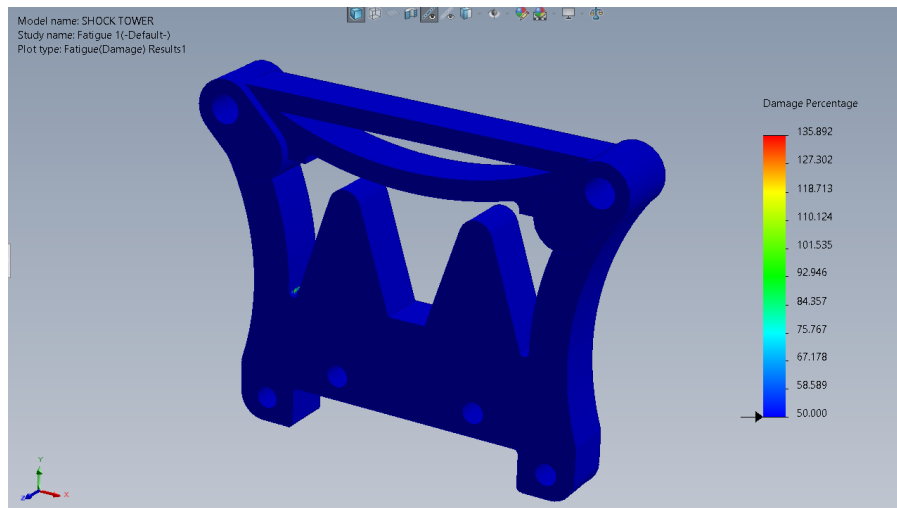


Figure 11. Fatigue damage distribution across the shock tower.

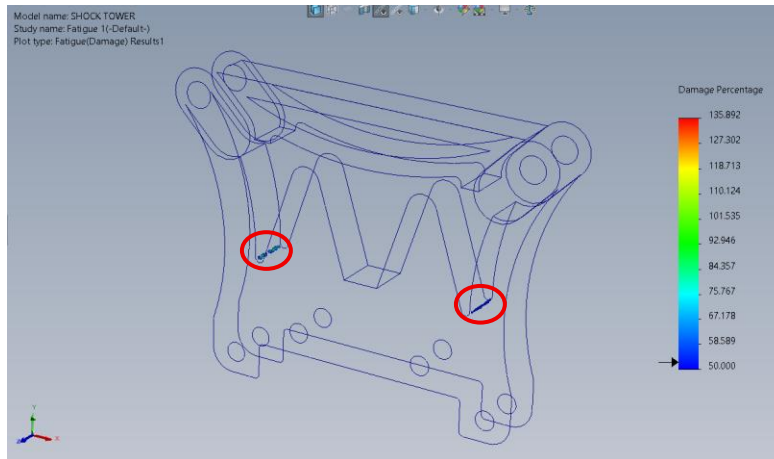


Figure 12. iso clipping for Fatigue damage distribution across the shock tower showing the most damaged regions.

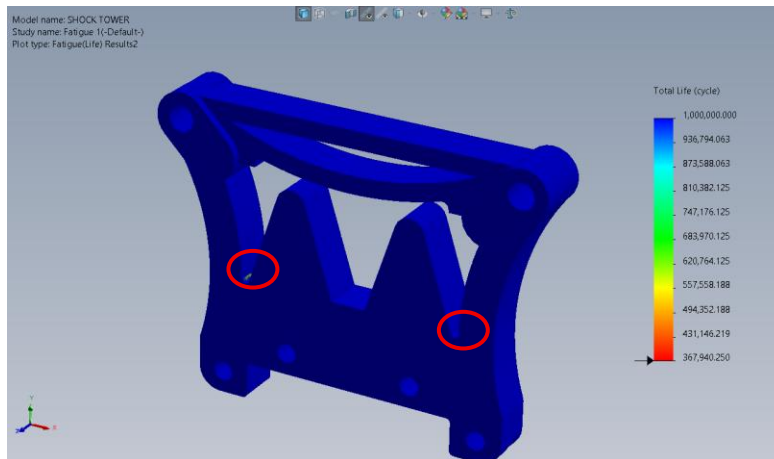


Figure 14. life cycle plot.

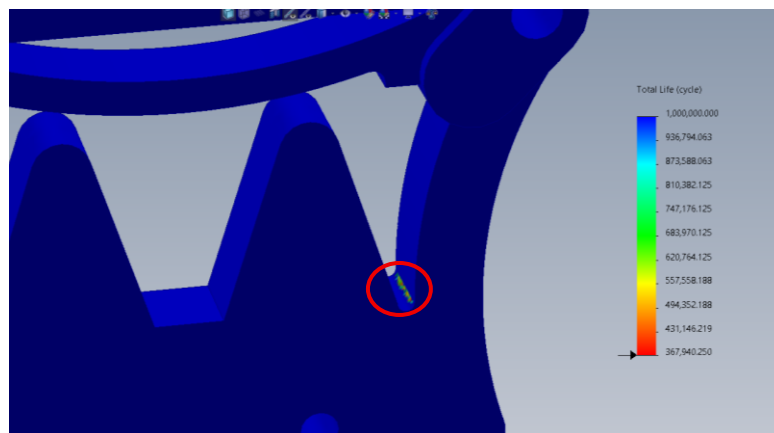


Figure 13. life cycle plot (zoomed into critical region).

Table 3. Fatigue Analysis result summery.

Life cycles	Percentage damage after 500,000 fully reversed loading
367,940	136 % at small specific locations

12. Topology Optimization

A topology optimization study was conducted to identify regions where material can be removed while maintaining stiffness and safety.

Optimization Objectives:

- Best stiffness-to-weight ratio
- Reduce mass by **30%**

Constraints:

- Minimum **Factor of Safety = 2**
- Preserved regions around all mounting pins

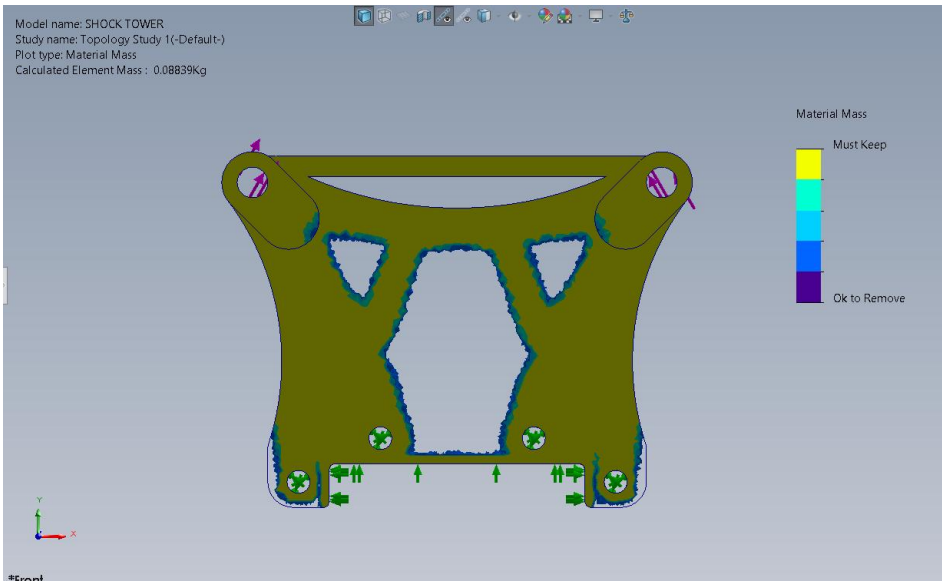


Figure 15. Topology optimization design space with preserved mounting regions (pins locations).

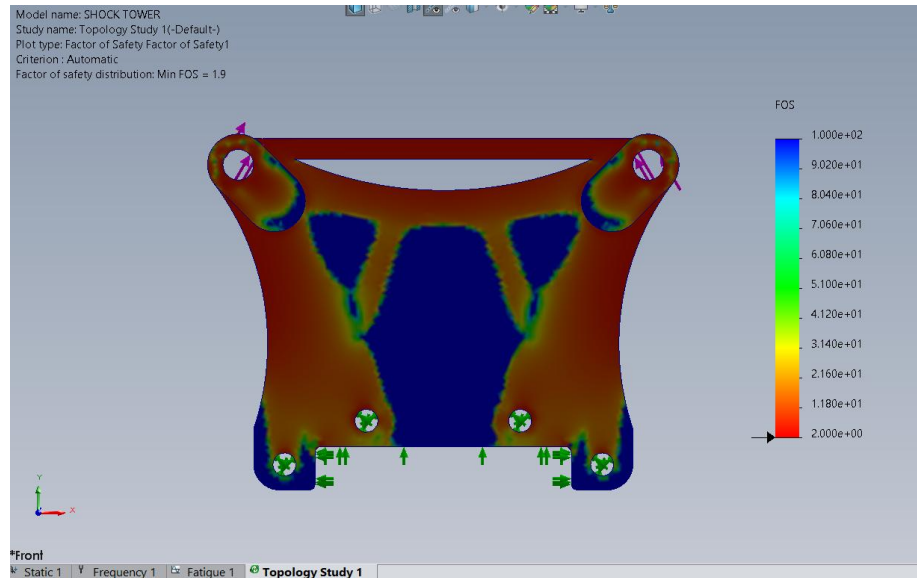


Figure 16. factor of safety distribution.

13. Key Results Summary

Study	Result
Static Analysis	Acceptable stress and displacement
Trend Tracker	Identified best configuration design
Modal Analysis	Identify the point representing the risk of resonance 2 nd & 3 rd modes
Fatigue Analysis	Survives 500,000 cycles
Topology Optimization	Potential mass reduction $\approx$ 30%

14. Conclusion

This project demonstrates a complete structural simulation workflow for evaluating and improving an RC shock tower design.

Static analysis was used to assess structural behavior under loading conditions, while modal analysis identified potential resonance risks. Fatigue analysis evaluated durability under cyclic loads, and topology optimization revealed opportunities for significant mass reduction while maintaining structural safety.